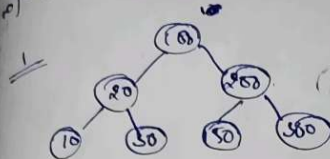


Q. 10/05/26

CO3 - Q93 Tree Traversal challenge



preorder (NLR) :-

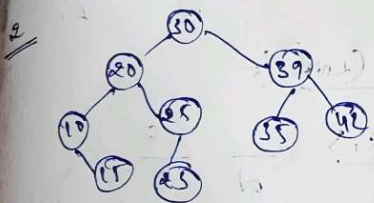
100
N
20, 10, 30
L
50, 30, 60
R

inorder (LNR) :-

10, 20, 30, 50, 60
L
100
N
150, 200, 300
R
1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100

post order (LRN) :-

10, 30, 20, 50, 60, 100
L
150, 30, 20, 100
R



preorder (NLR) :-

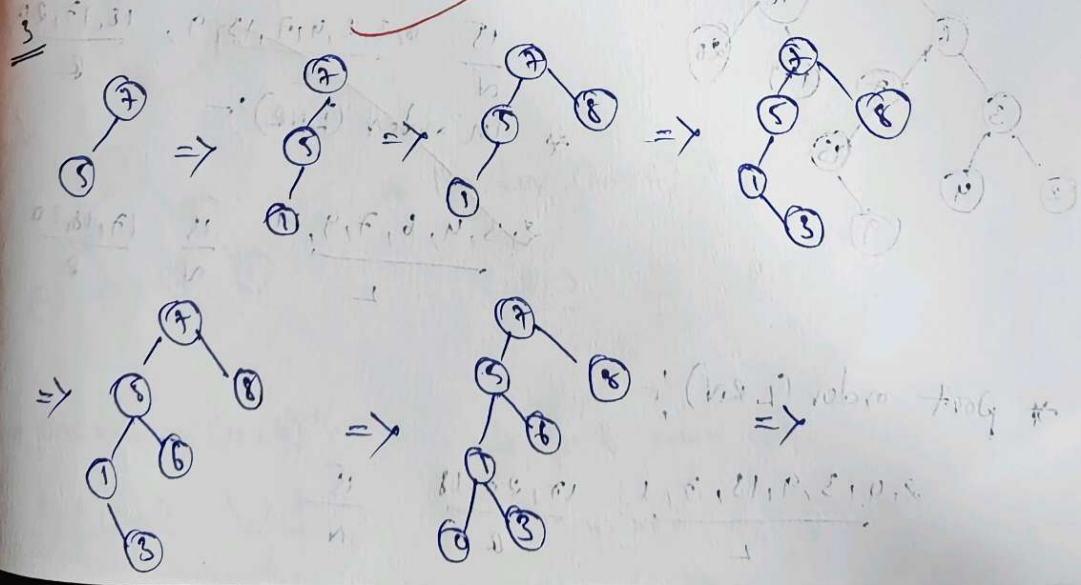
30
N
20, 10, 15, 35, 40
L
50, 35, 40
R

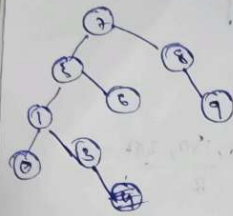
inorder (LNR) :-

10, 15, 20, 25, 28, 30, 35, 39, 42
L
30
N
35, 39, 42
R

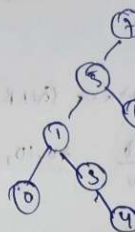
post order (LRN) :-

15, 10, 25, 25, 20, 20, 35, 42, 39, 30
L
30
N
35, 39, 42, 30
R





$\Rightarrow$



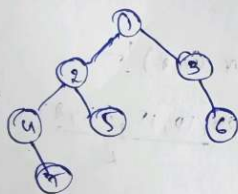
* Inorder (LNR) :-

0, 1, 3, 4, 5

$\frac{7}{N}$

$\frac{8, 9}{R}$

5



* preorder (NLR) :-

$\frac{1}{N}$

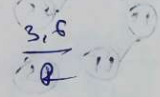
$\frac{2, 4, 7, 5}{L}$

$\frac{3, 6}{R}$

* Inorder (LNR) :-

2, 4, 7, 5

$\frac{1}{N}$



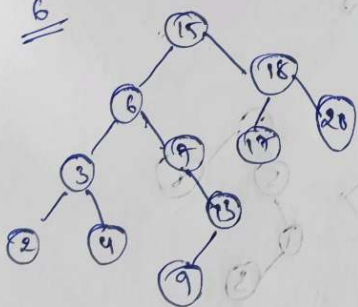
* post order (LRN) :-

2, 4, 5, 7

$\frac{3, 6}{R}$

$\frac{1}{N}$

6



* pre order (NLR) :-

$\frac{15}{N}$

$\frac{6, 3, 2, 4, 7, 13, 9}{L}$

$\frac{18, 17, 20}{R}$

* Inorder (LNR) :-

2, 3, 4, 6, 7, 9, 13

$\frac{15}{N}$

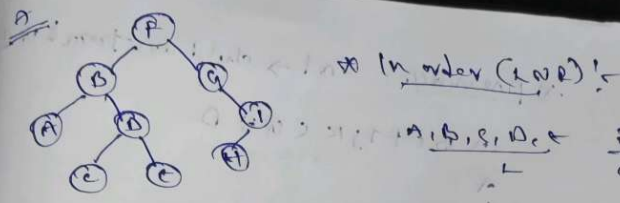
$\frac{17, 18, 20}{R}$

* post order (LRN) :-

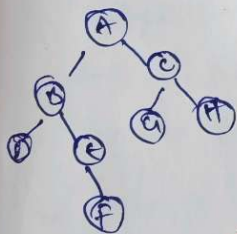
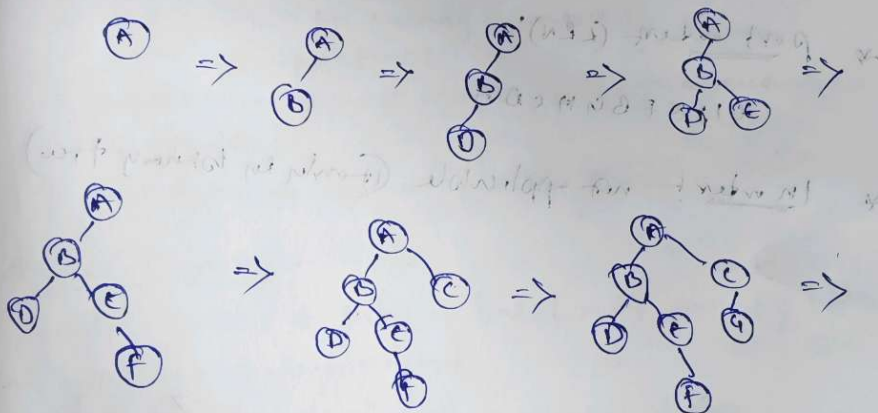
2, 4, 3, 9, 13, 7, 1

$\frac{17, 20, 18}{R}$

$\frac{15}{N}$

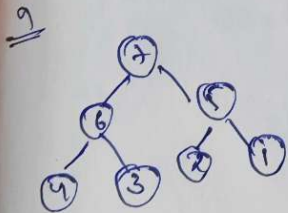


8. Given D B E F A G H C → inorder
A B D E F C H → preorder



* post order (RNL)!

$\frac{D, F, E, B}{L} \quad \frac{G, H, C}{R} \quad \frac{A}{2}$



* Inorder (LVR)!

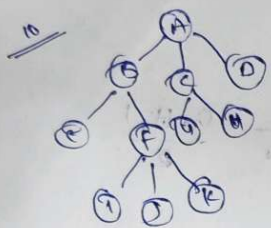
$\frac{4, 6, 3}{L} \quad \frac{7}{2} \quad \frac{2, 5, 1}{R}$

* preorder (NLR)!

$\frac{7, 6, 4, 3}{2} \quad \frac{5, 2, 1}{R}$

* post order (RNL)!

$\frac{4, 3, 6}{L} \quad \frac{2, 1, 5}{R} \quad \frac{7}{2}$



* preorder! Root $\rightarrow$ children from l to r

A B D E F J K C G H I

A
 $\downarrow$
 B $\rightarrow$ C $\rightarrow$ F $\rightarrow$ I $\rightarrow$ J $\rightarrow$ K

$\downarrow$
 C $\rightarrow$ G $\rightarrow$ H

$\downarrow$
 D

* post order (LRN):

E J K F B G H C D A

* In order! not applicable (only in binary tree)

Amal
 14/8/2020